

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 13A.1

EQUILIBRIUM CONSTANT, K_c

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A.1: Equilibrium Constant, K _c
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	8 Marks
EXAMINER FAQs:	3 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 13A.1

Q1. 1. (a) State the expression for the equilibrium constant, K_c , for the following homogeneous reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

[1 Marks]

WCH14/01/Jan23/Q3(a)

Q2. (b) A mixture of 2.0 mol of SO_2 and 1.5 mol of O_2 was placed in a 2.0 dm³ flask and allowed to reach equilibrium at a constant temperature. At equilibrium, 1.2 mol of SO_3 had formed. Calculate the value of K_c at this temperature and state its units.

[4 Marks]

WCH14/01/Jan23/Q3(b)

Q3. 2. What are the units of K_c for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$?

[1 Marks]

WCH14/01/Oct22/Q5

- A. mol dm⁻³
- B. dm³ mol⁻¹
- C. mol² dm⁻⁶
- D. dm⁶ mol⁻²

Q4. 3. Explain why the concentration of a solid is not included in the K_c expression for a heterogeneous equilibrium.

[2 Marks]

WCH14/01/Jun21/Q4(c)

SECTION B: WORKED MARK SCHEME — 13A.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) State the expression for the equilibrium constant, K_c , for the following homogeneous reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ $K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$ Allow correct expression without state symbols.	[1]
Q2	Q2: (b) A mixture of 2.0 mol of SO_2 and 1.5 mol of O_2 was placed in a 2.0 dm ³ flask and allowed to reach equilibrium at a constant temperature. At equilibrium, 1.2 mol of SO_3 had formed. Calculate the value of K_c at this temperature and state its units. 1. Moles of SO_3 at eqm = 1.2 mol 2. Moles of SO_2 at eqm = 2.0 - 1.2 = 0.8 mol 3. Moles of O_2 at eqm = 1.5 - 0.6 = 0.9 mol 4. Concentration: $[\text{SO}_3] = 0.6$, $[\text{SO}_2] = 0.4$, $[\text{O}_2] = 0.45$ mol dm ⁻³ 5. $K_c = \frac{(0.6)^2}{(0.4)^2 \times 0.45} = 5.0 \text{ dm}^3 \text{ mol}^{-1}$	[4]
Q3	Q3: 2. What are the units of K_c for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$? D	[1]
Q4	Q4: 3. Explain why the concentration of a solid is not included in the K_c expression for a heterogeneous equilibrium. 1. The concentration (or density) of a solid is constant. 2. It is incorporated into the value of the equilibrium constant K_c itself.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 13A.1**FAQ #1: Do I need to include units in my final answer for K_c ?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Do I need to include units in my final answer for K_c ?

MODEL ANSWER / EXAMINER EXPECTATION:

Yes, units are strictly required unless the calculation results in no units (e.g., when the moles on both sides cancel out). Always deduce the units by substituting mol dm^{-3} into the K_c expression.

FAQ #2: Why did I lose marks for my K_c expression?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why did I lose marks for my K_c expression?

MODEL ANSWER / EXAMINER EXPECTATION:

A common trap is using round brackets $()$ instead of square brackets $[]$. In chemistry, square brackets denote concentration in mol dm^{-3} , which is mandatory for K_c expressions. Another error is including solids in heterogeneous equilibria.

FAQ #3: How do I handle equilibrium calculations when volume is not given?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do I handle equilibrium calculations when volume is not given?

MODEL ANSWER / EXAMINER EXPECTATION:

If the number of moles of reactants and products are equal, the volume (V) will cancel out in the K_c expression. In such cases, you can use moles directly in the expression without calculating concentration.